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Linux Distribution Used	ICDFA KALI 2025.1 - ORACLE VM VIRTUAL BOX
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INTRODUCTION

In GRC, Linux files and Directory Management is useful in these ways:

Audit Trails: The whole essence of GRC is documentation of policies, logs, assessment and reports. Organized directories make all these easy to locate, retrieve and present during audits

Access Control: Linux commands allows for enforcement of least-privilege principles, directly supporting confidentiality requirements

Log Management: Frameworks in GRC require log retention and review. Knowing how to navigate, search and archive logs. Using linux commands like grep, tar, /var/log commands are essential for monitoring and incident response

Reproducibility and Integrity: Clear file structures and version control practices help demonstrate that records haven't been tampered with, supporting data integrity requirements.

Efficiency: Risk assessments and audits often involve large volumes of files; Scripting and command-line tools speed up sorting, filtering and reporting instead of manual reviews.

TASK 1 : Open the Shell and Identify the Environment

Commands Used -

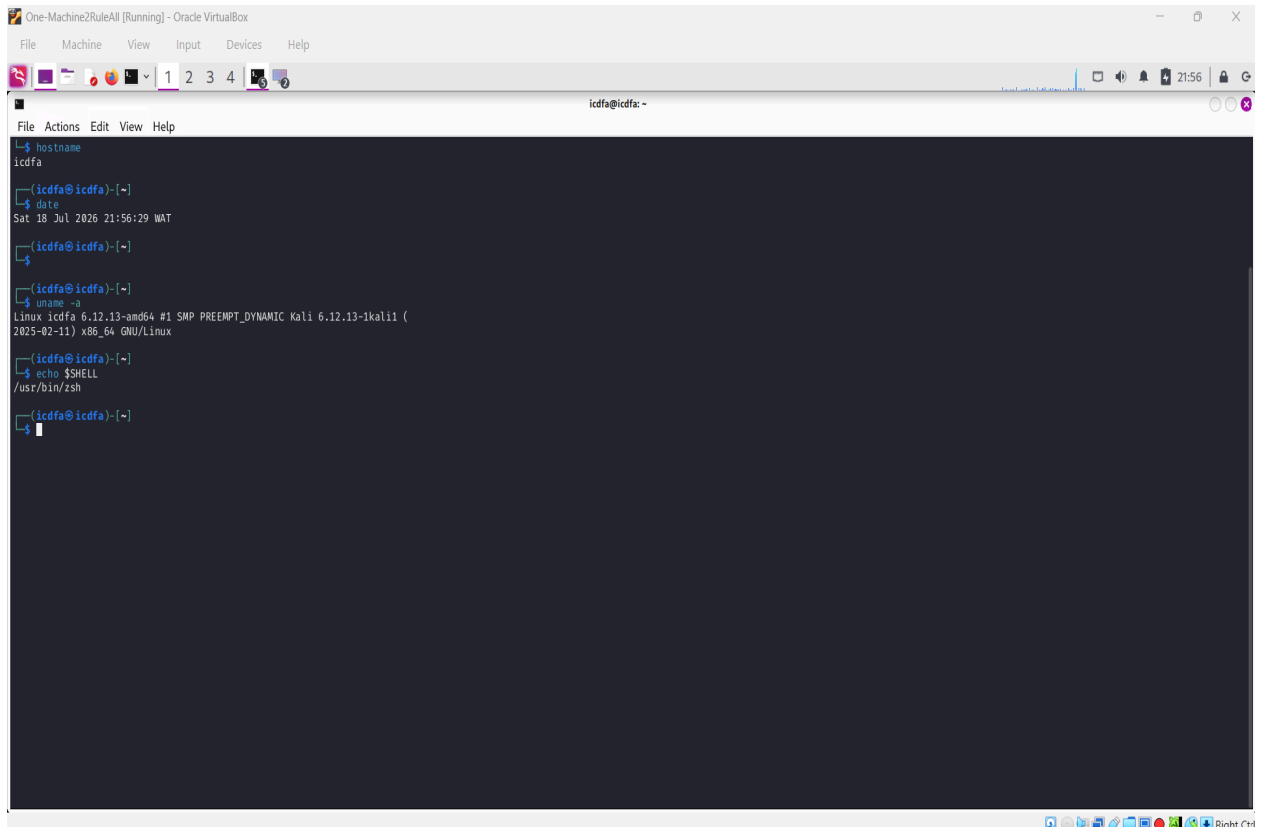
whoami,

hostname,

Date

uname - a

echo \$SHELL



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4 5 6
icdfa@icdfa: ~

icdfa@icdfa:~$ hostname
icdfa
icdfa@icdfa:~$ date
Sat 18 Jul 2026 21:56:29 WAT
icdfa@icdfa:~$
icdfa@icdfa:~$
icdfa@icdfa:~$ uname -a
Linux icdfa 6.12.13-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.12.13-1kali1 (
2025-02-11) x86_64 GNU/Linux
icdfa@icdfa:~$ echo $SHELL
/usr/bin/zsh
icdfa@icdfa:~$
```

These five commands give a quick snapshot of who you are, where you are, when it is, what system you're on, and how you're interacting with it, they are commonly chained together when troubleshooting or documenting a system's state.

- Linux username- **ICDFA**
- The computer or VM hostname- **ICDFA**
- The current date and time-**18TH JULY, 2026 , 21:56:29 WAT**
- The operating system/kernel information- **Linux icdfa 6.12.13-amd64 #1 SMP PREEMPT_DYNAMIC kali 6.12.13-1kali1 (2025-02-11) x86_64 GNU/Linux**
- The shell currently in use- **/usr/bin/zsh**

TASK 2: Create the Main GRC101 Laboratory Workspace

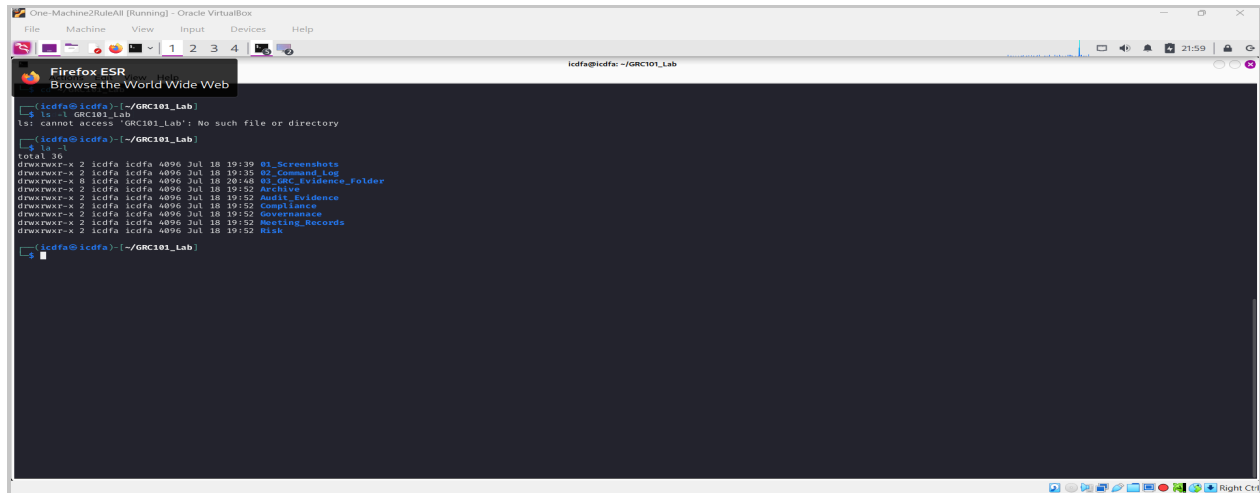
Commands used:

```
cd ~
```

```
mkdir -p GRC101_Lab/{01_Screenshots,02_Command_Log,03_GRC_Evidence_Folder}
```

```
ls -l GRC101_Lab
```

This is used to create a folder with sub-folders in one step and also to list the content on the folder as a confirmation that they were created



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdffa@icdffa: ~/GRC101_Lab

icdffa@icdffa:~/GRC101_Lab
$ ls -l GRC101_Lab
ls: cannot access 'GRC101_Lab': No such file or directory

icdffa@icdffa:~/GRC101_Lab
$ ls -l
total 16
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:39 01_Screenshots
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:35 02_Command_Log
drwxrwxr-x 8 icdffa icdffa 4096 Jul 18 20:48 03_GRC_Evidence_Folder
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Archive
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Audit_Evidence
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Compliance
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Governance
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Meeting_Records
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:52 Risk

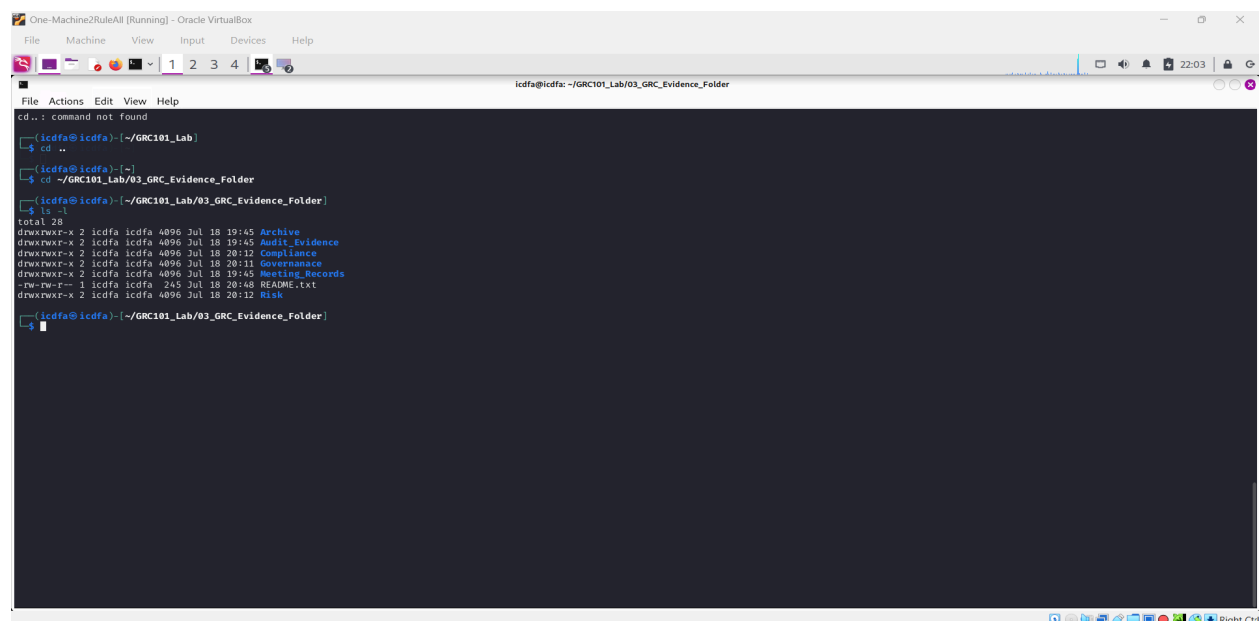
icdffa@icdffa:~/GRC101_Lab
$
```

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
```

```
mkdir -p Governance Risk Compliance Audit_Evidence Meeting_Records Archive
```

```
ls -l
```

This command moves the directory, creates six subfolders there and lists the content to confirm they were created



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdffa@icdffa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdffa@icdffa:~/GRC101_Lab
$ cd ..

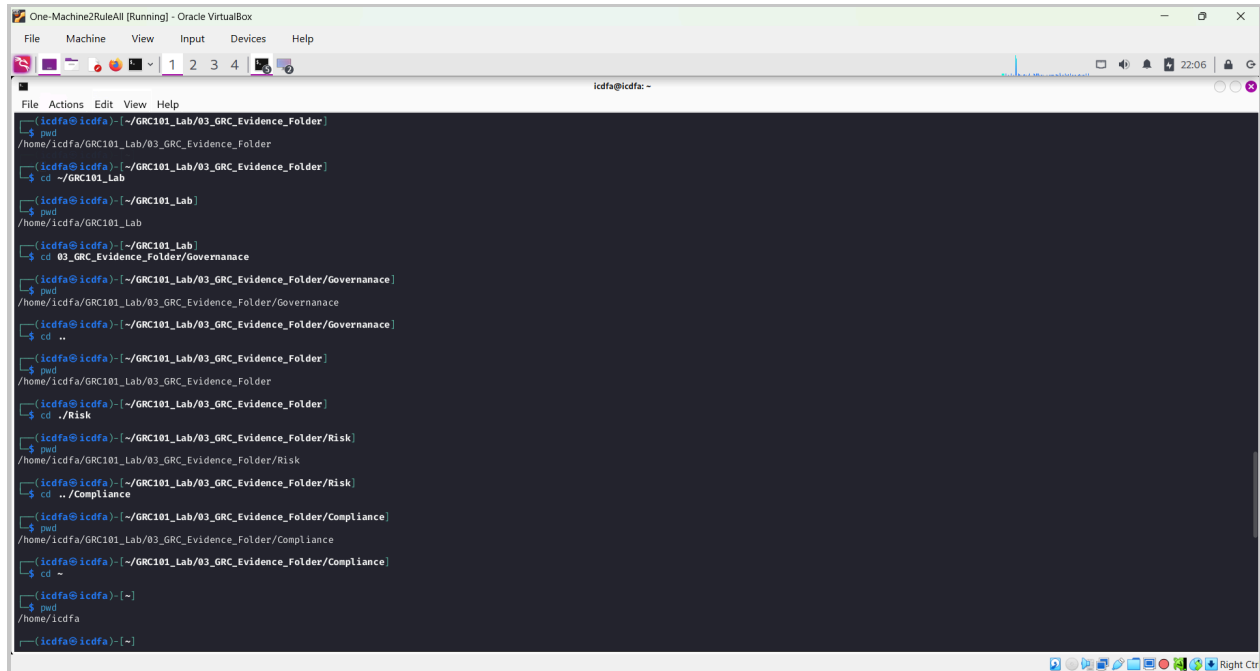
icdffa@icdffa:~/GRC101_Lab
$ cd ~/GRC101_Lab/03_GRC_Evidence_Folder

icdffa@icdffa:~/GRC101_Lab/03_GRC_Evidence_Folder
$ ls -l
total 28
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 20:12 Compliance
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 20:11 Governance
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdffa icdffa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdffa icdffa 4096 Jul 18 20:12 Risk

icdffa@icdffa:~/GRC101_Lab/03_GRC_Evidence_Folder
$
```

TASK 3: Navigate the Linux File System

Commands used `pwd , cd ~/GRC101_Lab ,pwd ,cd 03_GRC_Evidence_Folder/Governance ,pwd ,cd .. ,pwd ,cd ,./Risk ,pwd ,cd ../Compliance ,pwd ,cd ~ ,pwd`



```
One-Machine2RuleAll [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~
File Actions Edit View Help
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa: ~
$ cd ~/GRC101_Lab
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab
icdfa@icdfa: ~
$ cd 03_GRC_Evidence_Folder/Governance
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder/Governance
icdfa@icdfa: ~
$ cd ..
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa: ~
$ cd ./Risk
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder/Risk
icdfa@icdfa: ~
$ cd ../Compliance
icdfa@icdfa: ~
$ pwd
/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder/Compliance
icdfa@icdfa: ~
$ cd ~
icdfa@icdfa: ~
$ pwd
/home/icdfa
icdfa@icdfa: ~
```

These commands print the current directory (`pwd`) at each step while navigating: from wherever you start, into `~/GRC101\_Lab`, down into `03\_GRC\_Evidence\_Folder/Governance`, back up one level (`..`), then attempts `cd` (which jumps home) followed by `./Risk` as a separate, likely broken command since `cd` and `./Risk` need a comma-separated command boundary, not shown correctly. The rest moves up into `Compliance` from wherever you landed, then finally back to your home directory (`~`), printing the path after each move.

- What pwd does-prints the current directory (`pwd`) at each step while navigating: from wherever you start
- The difference between `cd ..` and `cd ~`. `Cd` takes you home while `cd ~` takes you back to the home directory
- The meaning of `.` and `..` - “.” takes you back one level and “..” takes you back 2 levels
- The difference between an absolute path and a relative path-

Absolute path: starts from the root of the filesystem (or `~` for home) and always points to the same location no matter where you currently are — e.g. `/home/user/GRC101_Lab/Governance`` or `~/GRC101_Lab/Governance``.

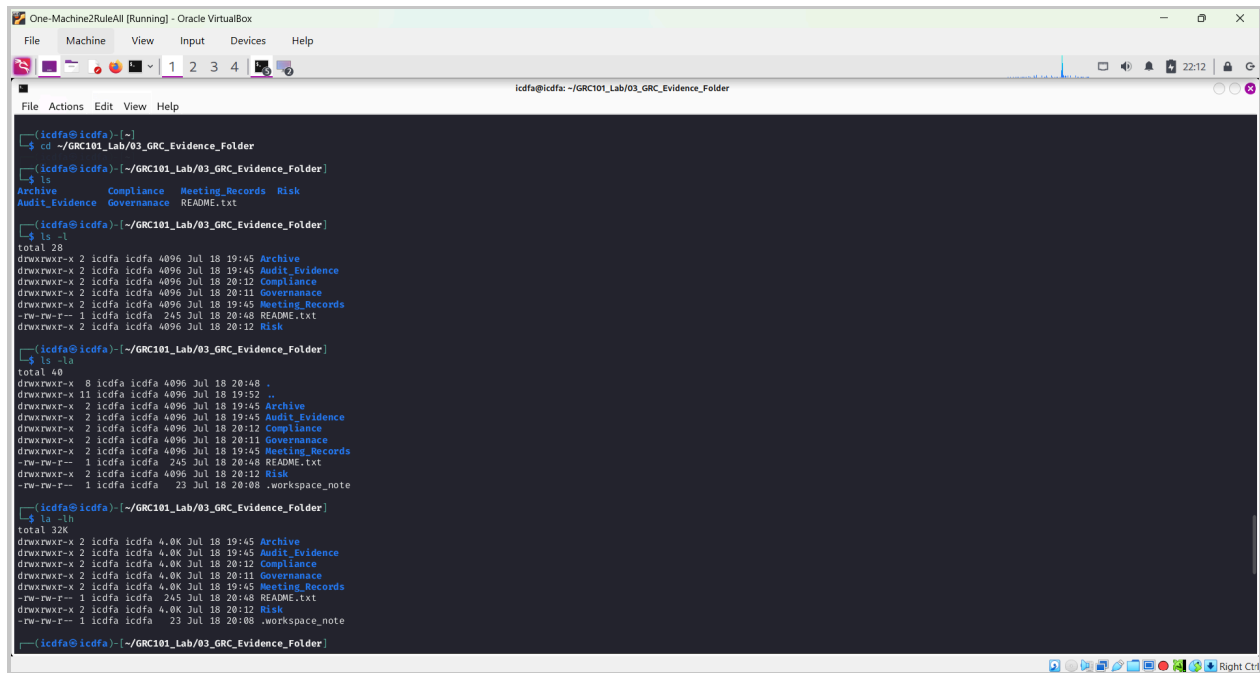
Relative path: starts from your *current* directory, so it depends on where you are when you use it e.g. ``Governance`` or `../Compliance``.

TASK 4: List and Inspect Files and Directories

Commands used

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
```

```
ls, ls -l, ls -la, ls -lh
```



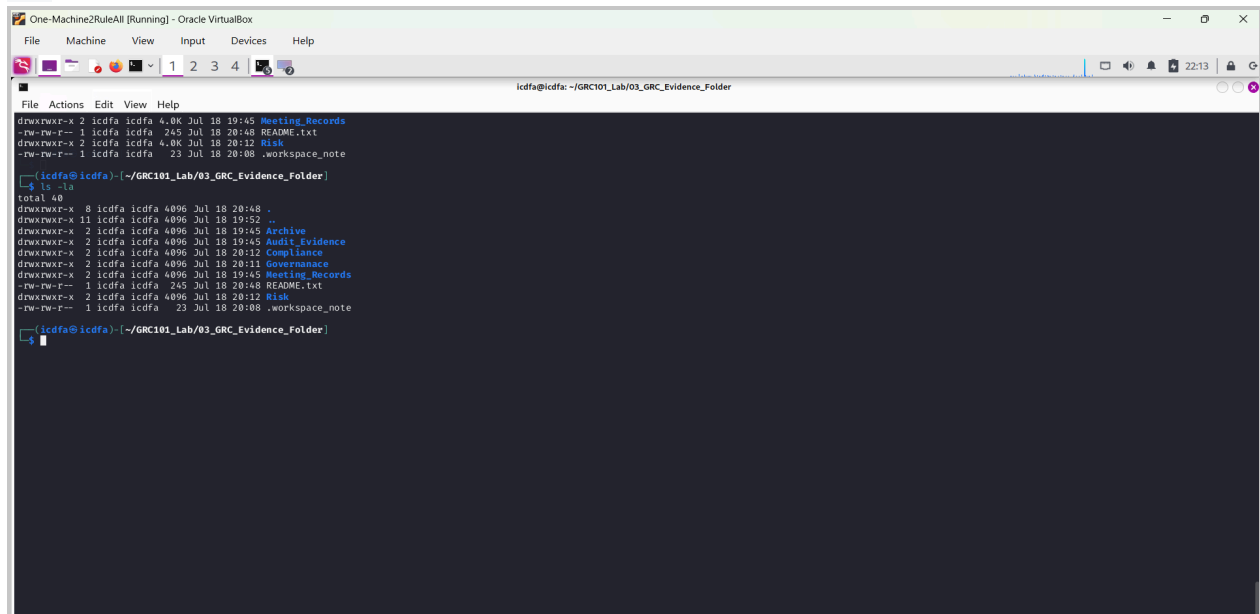
```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdfa@icdfa:~$ cd ~/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls
Archive      Compliance  Meeting_Records  Risk
Audit_Evidence  Governance  README.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -l
total 28
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Compliance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:11 Governance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Risk
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -la
total 40
drwxrwxr-x 8 icdfa icdfa 4096 Jul 18 20:48 .
drwxrwxr-x 11 icdfa icdfa 4096 Jul 18 19:52 ..
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Compliance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:11 Governance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Risk
-rw-rw-r-- 1 icdfa icdfa 23 Jul 18 20:08 .workspace_note
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -lh
total 22K
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 20:12 Compliance
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 20:11 Governance
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 20:12 Risk
-rw-rw-r-- 1 icdfa icdfa 23 Jul 18 20:08 .workspace_note
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
```

```
echo 'Internal GRC workspace' > .workspace_note
```

```
ls -la
```



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4.0K Jul 18 20:12 Risk
-rw-rw-r-- 1 icdfa icdfa 23 Jul 18 20:08 .workspace_note
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -la
total 40
drwxrwxr-x 8 icdfa icdfa 4096 Jul 18 20:48 .
drwxrwxr-x 11 icdfa icdfa 4096 Jul 18 19:52 ..
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Compliance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:11 Governance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Risk
-rw-rw-r-- 1 icdfa icdfa 23 Jul 18 20:08 .workspace_note
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
```

- Which command displays hidden files? ANS `ls -la`
- What information is shown by `ls -l`?

The command shows a detailed listing with these columns for each file/folder:

- Permissions** (e.g. `drwxr-xr-x`) : type and read/write/execute rights
- Number of links
- Owner username
- Group name
- Size in bytes
- Last modified date/time
- Name of the file or folder

- Why might a GRC team use hidden configuration files?

1. They store settings for tools (audit scripts, compliance scanners, logging agents) that don't need to be seen or edited during normal daily use
2. Keeping them hidden reduces clutter and the chance of accidental deletion or modification, which matters for maintaining audit integrity and evidence trails

TASK 5: Create and View GRC Files

Create simple GRC documents using touch, echo and cat:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
```

```
touch Governance/information_security_policy.txt
```

```
touch Risk/risk_register.txt
```

```
touch Compliance/compliance_checklist.txt
```

```
echo "NorthStar Services Ltd - Information Security Policy" > Governance/information_security_policy.txt
```

```
echo "Owner: GRC Department" >> Governance/information_security_policy.txt
```

```
echo "Review Cycle: Annual" >> Governance/information_security_policy.txt
```

```
echo "Risk ID | Risk Description | Owner | Status" > Risk/risk_register.txt
```

```
echo "R-001 | Unauthorised access to records | IT Manager | Open" >> Risk/risk_register.txt
```

```
echo "Control | Requirement | Status" > Compliance/compliance_checklist.txt
```

```
echo "Access Review | User access reviewed quarterly | Pending" >> Compliance/compliance_checklist.txt
```

```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdfa@icdfa:~$ cd ~/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ touch Governance/information_sec_policy.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ touch Risk/risk_reg.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ touch Compliance/compliance_check.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "NorthStar Servi Ltd - Information Sec Policy">Governance/information_sec_policy.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "Owner: GRC Department"> Governance/information_sec_policy.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "Review Cycle: Annual"> Governance/information_sec_policy.txt
zsh: no such file or directory: Governance/information_sec_policy.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "Risk ID | Risk Description | Owner | Status">Risk/risk_reg.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "R-001 | Unauthorised access to records | IT Manager | Open">Risk/risk_reg.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ echo "Control | Requirement | Status">Compliance/compliance_check.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
```

Display the contents:

```
cat Governance/information_security_policy.txt
cat Risk/risk_register.txt
cat Compliance/compliance_checklist.txt
head -n 2 Governance/information_security_policy.txt
```

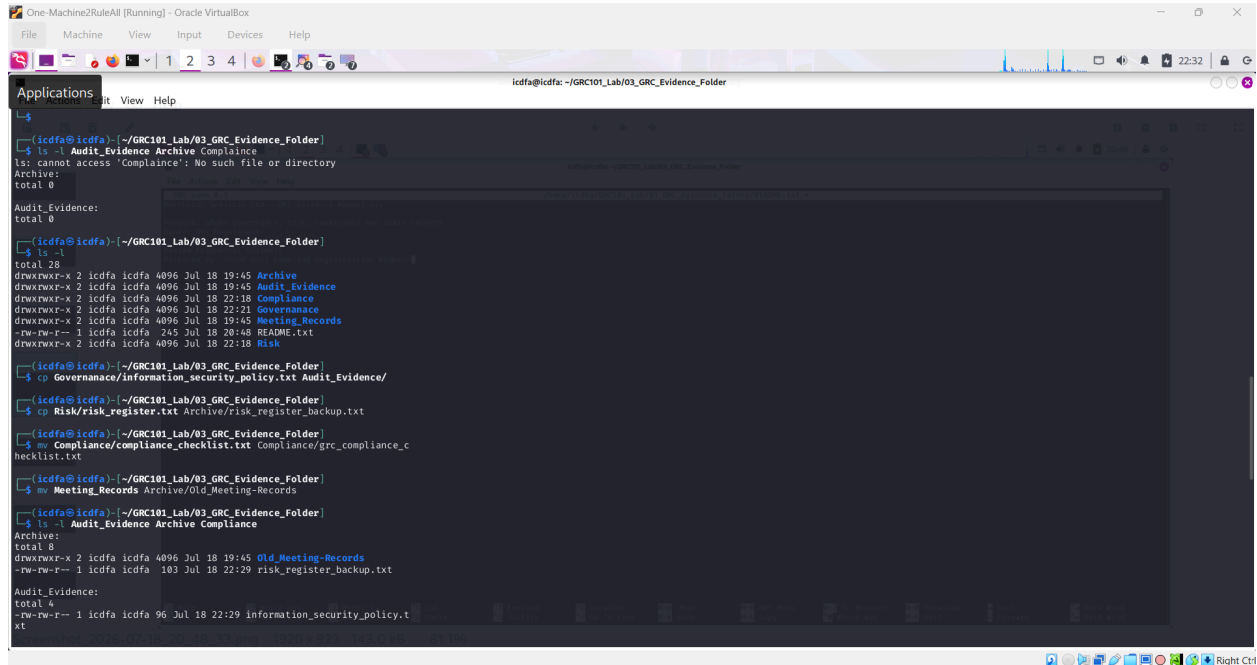
```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdfa@icdfa:~$ cd ~/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ cat Governance/information_security_policy.txt
NorthStar Services Ltd - Information Security Policy
Owner: GRC Department
Review Cycle: Annual
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ cat Risk/risk_register.txt
Risk ID | Risk Description | Owner | Status
R-001 | Unauthorised access to records | IT Manager | Open
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ cat Compliance/compliance_checklist.txt
cat: Compliance/compliance_checklist.txt: No such file or directory
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ head -n 2 Governance/information_security_policy.txt
head: cannot open 'Governance/information_security_policy.txt' for reading: No such file or directory
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
```

TASK 6 : Copy, Move and Rename GRC Records

Use cp and mv to manage copies of documents:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
cp Governance/information_security_policy.txt Audit_Evidence/
cp Risk/risk_register.txt Archive/risk_register_backup.txt
mv Compliance/compliance_checklist.txt Compliance/grc_compliance_checklist.txt
mv Meeting_Records Archive/Old_Meeting_Records
ls -l Audit_Evidence Archive Compliance
```



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

Applications
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -l Audit_Evidence Archive Compliance
ls: cannot access 'Compliance': No such file or directory
Archive:
total 0
Audit_Evidence:
total 0
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -l
total 28
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 22:18 Compliance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 22:21 Governance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 245 Jul 18 20:48 README.txt
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 22:18 Risk
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ cp Governance/information_security_policy.txt Audit_Evidence/
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ cp Risk/risk_register.txt Archive/risk_register_backup.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ mv Compliance/compliance_checklist.txt Compliance/grc_compliance_c
hecklist.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ mv Meeting_Records Archive/Old_Meeting_Records
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ ls -l Audit_Evidence Archive Compliance
Archive:
total 8
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Old_Meeting_Records
-rw-rw-r-- 1 icdfa icdfa 103 Jul 18 22:29 risk_register_backup.txt
Audit_Evidence:
total 4
-rw-rw-r-- 1 icdfa icdfa 96 Jul 18 22:29 information_security_policy.t
xt
```

Explain the difference between cp and mv. State which command you would use to keep an original file while creating a duplicate for an auditor.

ANSWER

cp copies a file, leaving the original in place and creating a duplicate.

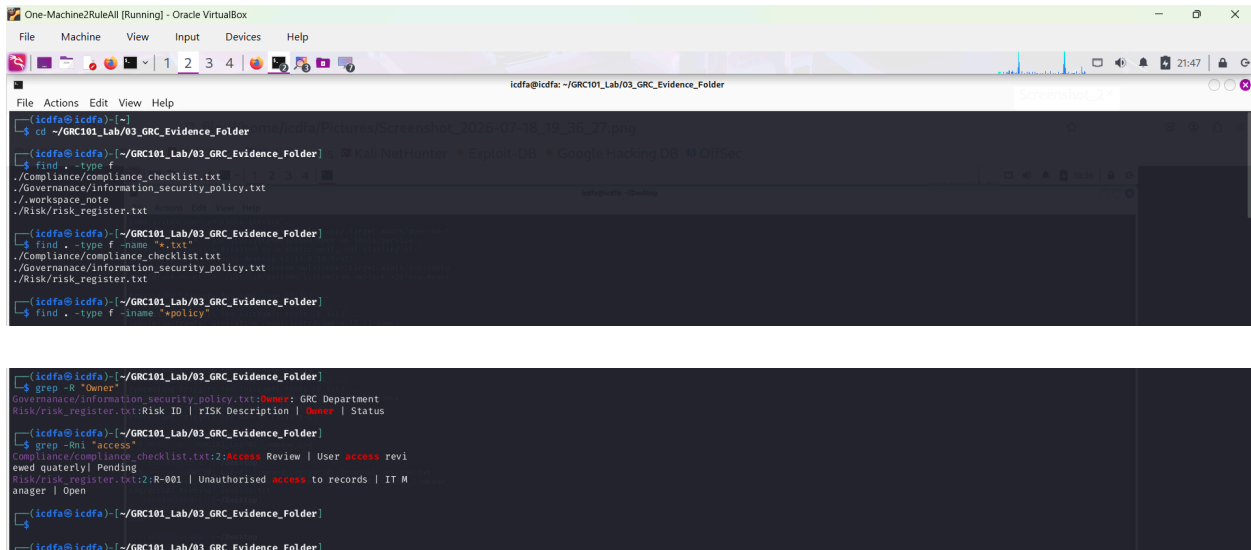
mv moves (or renames) a file, so the original location no longer has it; nothing is duplicated.

To keep the original while giving an auditor a duplicate, use `cp`

TASK 7: Search for Files and Information

Use find and grep to locate records and text:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
find . -type f
find . -type f -name "*.txt"
find . -type f -iname "*policy*"
grep -R "Owner" .
grep -Rni "access" .
```



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdfa@icdfa:~$ cd ~/GRC101_Lab/03_GRC_Evidence_Folder
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ find . -type f
./Compliance/compliance_checklist.txt
./Governance/information_security_policy.txt
./workspace_note
./Risk/risk_register.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ find . -type f -name "*.txt"
./Compliance/compliance_checklist.txt
./Governance/information_security_policy.txt
./Risk/risk_register.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ find . -type f -iname "*policy*"
./Governance/information_security_policy.txt
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ grep -R "Owner"
Governance/information_security_policy.txt:Owner: GRC Department
Risk/risk_register.txt:Risk ID | RISK Description | Owner | Status
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$ grep -Rni "access"
Compliance/compliance_checklist.txt:2:Access Review | User access reviewed quarterly | Pending
Risk/risk_register.txt:2:R-001 | Unauthorised access to records | IT Manager | Open
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder$
```

TASK 8 : View File Metadata and Inode Numbers

Inspect the files using long listing and inode options:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
ls -li Governance/information_security_policy.txt
stat Governance/information_security_policy.txt
df -h
df -i
```

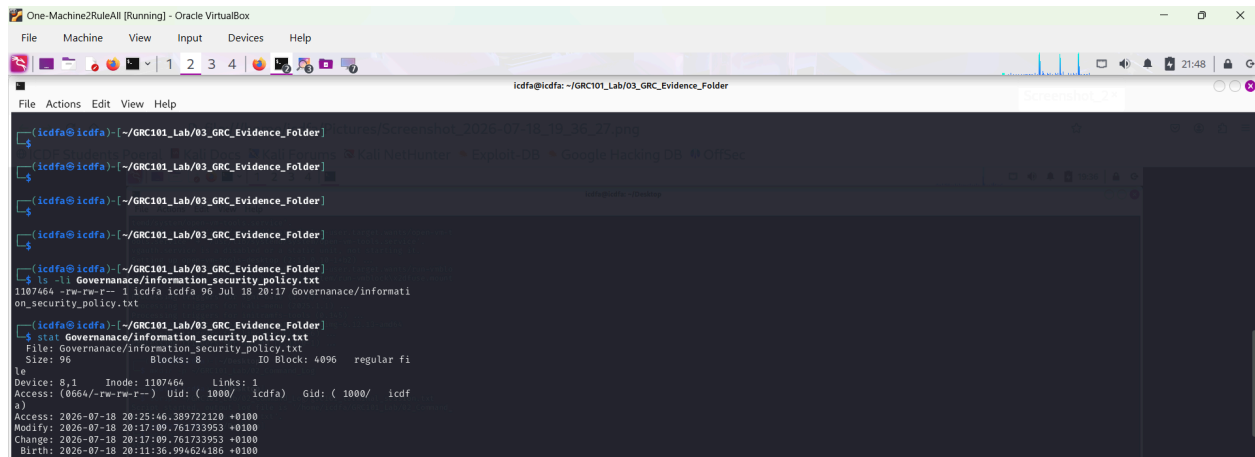
In your report, identify the inode number, owner, file size and last modification time of the policy file.

Inode number: 1107464

- Owner: icdfa (UID 1000, GID 1000/icdfa)
- File size: 96 bytes
- Last modification time: 2026-07-18 20:17:09.761733953 +0100

Briefly explain why file metadata can support audit evidence.

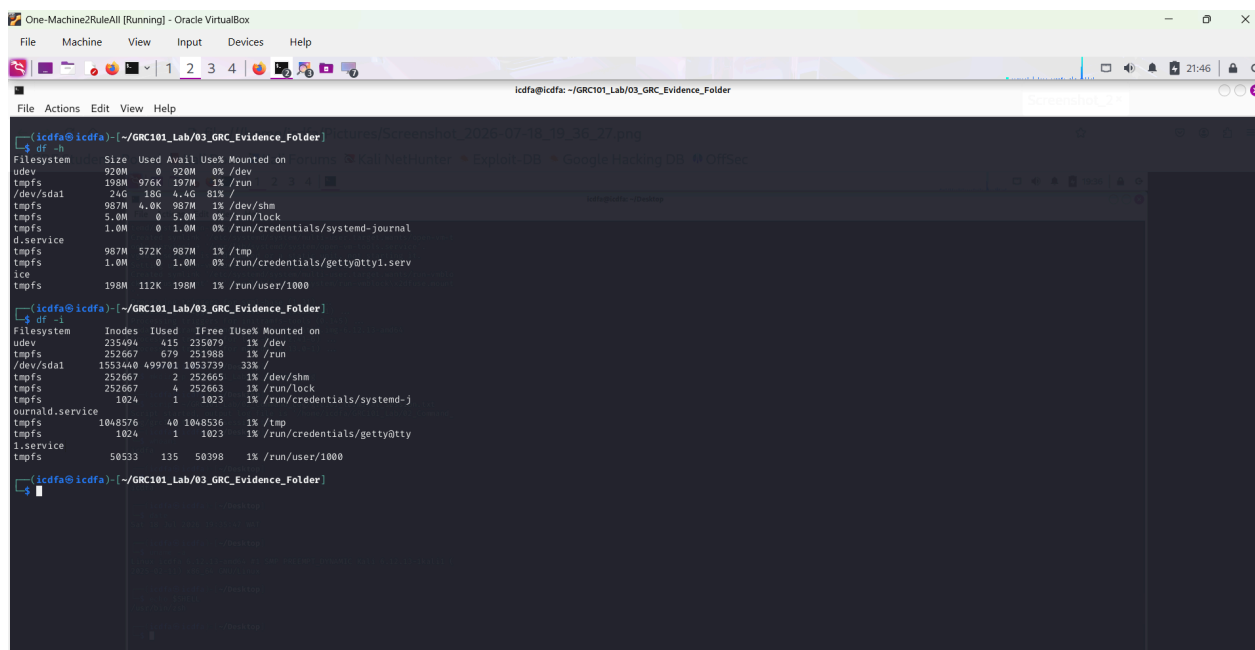
ANS- File metadata like timestamps, ownership, and inode data provides an independent, system-generated record of who created or modified a file and when, making it harder to falsify and useful for verifying the integrity and authenticity of audit evidence.



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

(icdfa@icdfa) ~/GRC101_Lab/03_GRC_Evidence_Folder
$ ls -l Governance/information_security_policy.txt
-rw-rw-r-- 1 icdfa icdfa 96 Jul 18 20:17 Governance/information_security_policy.txt

(icdfa@icdfa) ~/GRC101_Lab/03_GRC_Evidence_Folder
$ stat Governance/information_security_policy.txt
  File: Governance/information_security_policy.txt
  Size: 96          Blocks: 8          IO Block: 4096   regular file
Device: 8,1      Inode: 1107464     Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/  icdfa)   Gid: ( 1000/  icdfa)
Access: 2026-07-18 20:25:40.389721200 +0100
Modify: 2026-07-18 20:17:09.761733953 +0100
Change: 2026-07-18 20:17:09.761733953 +0100
 Birth: 2026-07-18 20:11:36.994624186 +0100
```



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

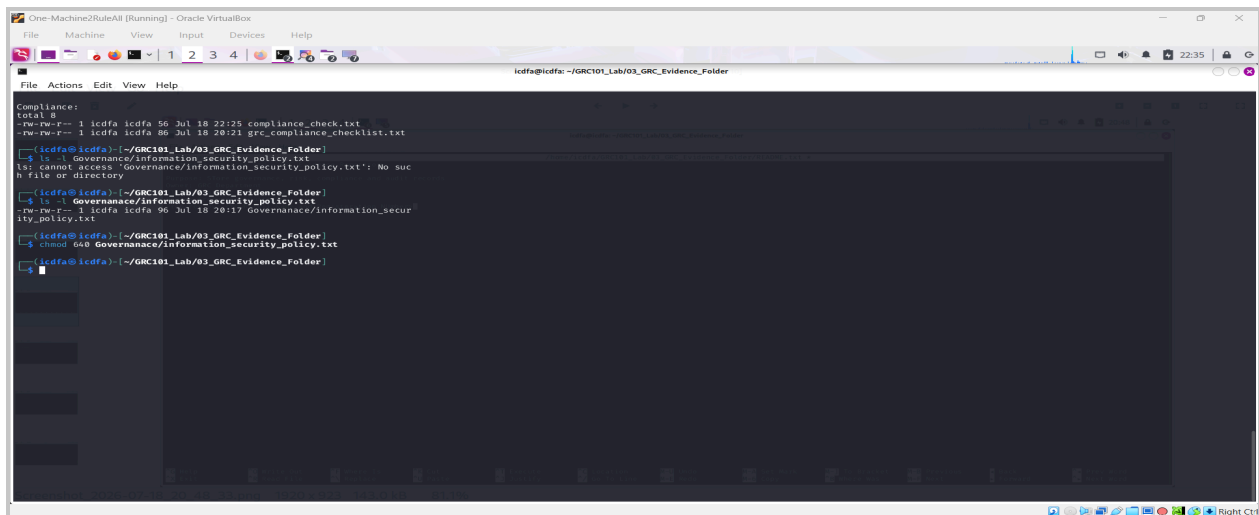
(icdfa@icdfa) ~/GRC101_Lab/03_GRC_Evidence_Folder
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            920M    0  920M   0% /dev
tmpfs           198M   976K  197M   1% /run
/dev/sda1       24G   18G  4.4G  81% /
tmpfs           987M   4.0K  987M   1% /dev/shm
tmpfs           5.0M    0  5.0M   0% /run/lock
tmpfs           1.0M    0  1.0M   0% /run/credentials/systemd-journal
d.service       987M  572K  987M   1% /tmp
tmpfs           1.0M    0  1.0M   0% /run/credentials/getty@tty1.serv
ice             198M  112K  198M   1% /run/user/1000

(icdfa@icdfa) ~/GRC101_Lab/03_GRC_Evidence_Folder
$ df -li
Filesystem      Inodes  IUsed  IFree IUse% Mounted on
udev            235494    415 235079   1% /dev
tmpfs           252667    679 251988   1% /run
/dev/sda1      1552440 499701 1052739  33% /
tmpfs           252667     2 252665   1% /dev/shm
tmpfs           252667     4 252663   1% /run/lock
tmpfs           1024      1  1023   1% /run/credentials/systemd-j
ournal.service
tmpfs          1048576    40 1048536   1% /tmp
tmpfs           1024      1  1023   1% /run/credentials/getty@tty
1.service
tmpfs           50533    135 50398   1% /run/user/1000
```

TASK 9: Basic Permissions and Responsible Use of sudo

Check and change permissions only on the practice policy file:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
ls -l Governance/information_security_policy.txt
chmod 640 Governance/information_security_policy.txt
ls -l Governance/information_security_policy.txt
```



```
One-Machine2RuleAll (Running) - Oracle VM VirtualBox
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

Compliance:
total 8
-rw-rw-r-- 1 icdfa icdfa 56 Jul 18 22:25 compliance_check.txt
-rw-rw-r-- 1 icdfa icdfa 86 Jul 18 20:21 grc-compliance-checklist.txt

icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder
$ ls -l Governance/information_security_policy.txt
ls: cannot access 'Governance/information_security_policy.txt': No such file or directory

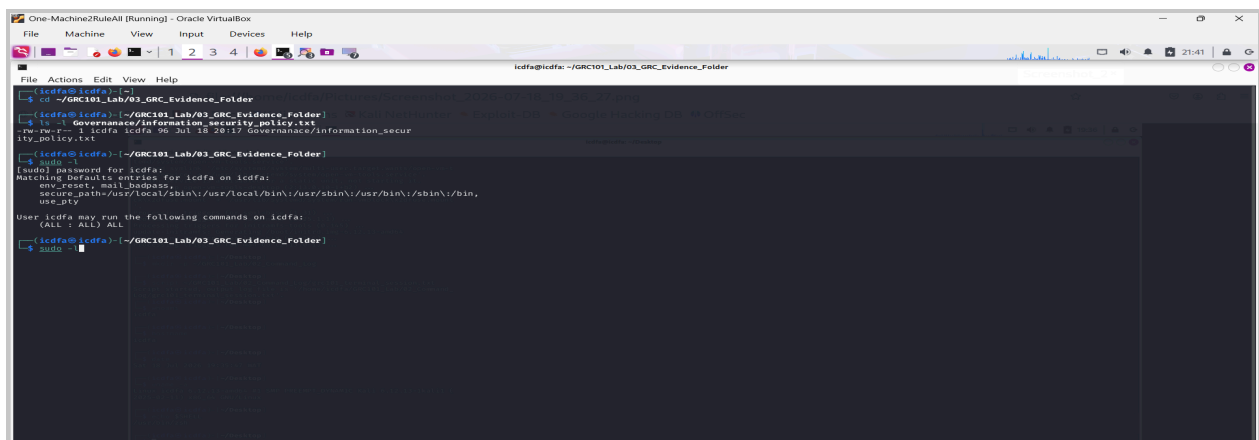
icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder
$ ls -l Governance/information_security_policy.txt
ls: cannot access 'Governance/information_security_policy.txt': No such file or directory

icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder
$ chmod 640 Governance/information_security_policy.txt
chmod: cannot access 'Governance/information_security_policy.txt': No such file or directory

icdfa@icdfa:~/GRC101_Lab/03_GRC_Evidence_Folder
$
```

Check whether sudo is available without making any system change:

```
sudo -l
```



```
One-Machine2RuleAll (Running) - Oracle VM VirtualBox
icdfa@icdfa: ~/GRC101_Lab/03_GRC_Evidence_Folder

icdfa@icdfa:~$ sudo -l
[sudo] password for icdfa:
Matching Defaults entries for icdfa on icdfa:
env_reset, mail_badpass,
secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin,
use_pty

User icdfa may run the following commands on icdfa:
(All ! ALL) ALL

icdfa@icdfa:~$ sudo
```

Explain what the owner, group and others can do after chmod 640.

After `chmod 640`:

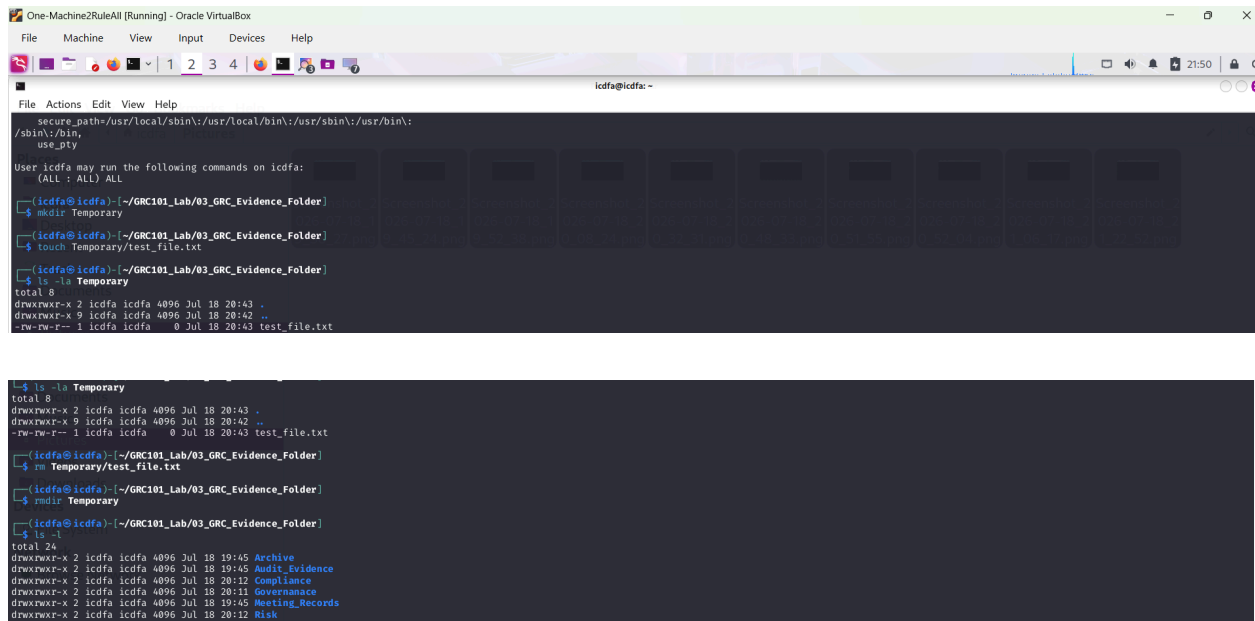
A Owner:- read + write (can view and edit the file)

- B Group:- read only (can view but not edit)
C Others:- no permissions (cannot read, write, or execute the file)

TASK 10: Safely Delete Practice Files and Empty Directories

Create temporary practice items and delete only those items:

```
cd ~/GRC101_Lab/03_GRC_Evidence_Folder
mkdir Temporary
touch Temporary/test_file.txt
ls -la Temporary
rm Temporary/test_file.txt
rmdir Temporary
ls -l
```



```
One-Machine2RuleAll [Running] - Oracle VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~
secure_path=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/usr/sbin:/bin:/usr/bin
/bin:/usr/bin:/usr/sbin:/usr/local/bin:/usr/local/sbin:
use_ptty
User icdfa may run the following commands on icdfa:
(ALL : ALL) ALL
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ mkdir Temporary
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ touch Temporary/test_file.txt
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ ls -la Temporary
total 8
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:43 .
drwxrwxr-x 9 icdfa icdfa 4096 Jul 18 20:42 ..
-rw-rw-r-- 1 icdfa icdfa 0 Jul 18 20:43 test_file.txt
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ rm Temporary/test_file.txt
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ rmdir Temporary
(icdfa@icdfa) [~/GRC101_Lab/03_GRC_Evidence_Folder]
$ ls -l
total 24
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Archive
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Audit_Evidence
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Compliance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:11 Governance
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 19:45 Meeting_Records
drwxrwxr-x 2 icdfa icdfa 4096 Jul 18 20:12 Risk
```

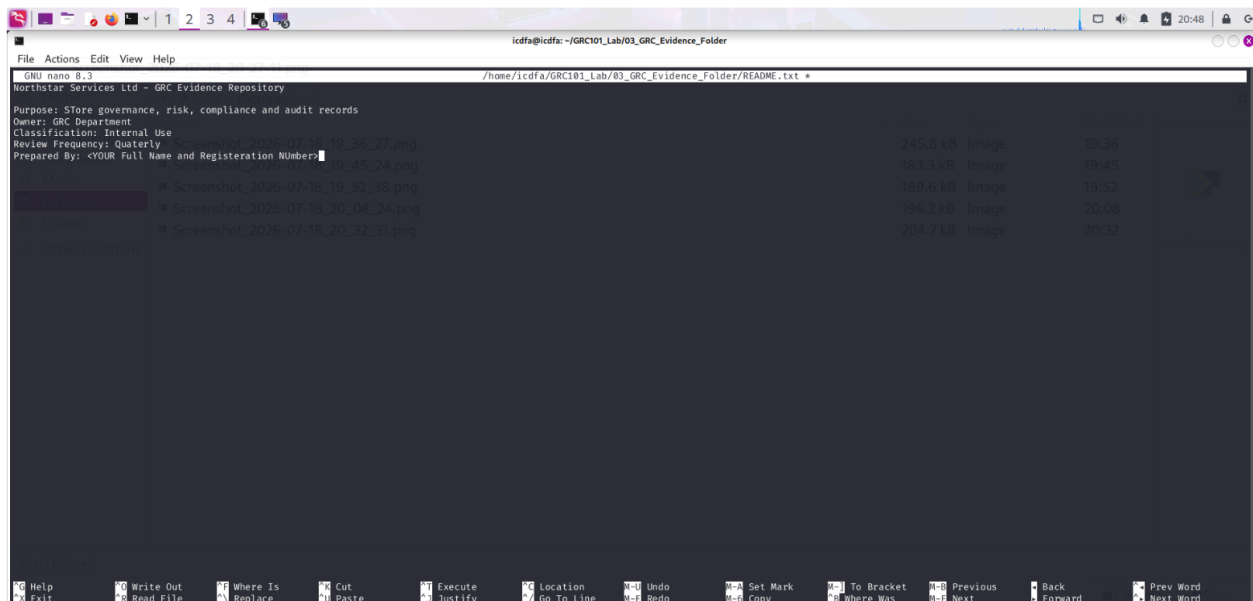
Explain why rmdir works only when a directory is empty:-

`rmdir` only removes empty directories because it's designed as a safety check to prevent AN erroneous deletion of a directory's contents; to remove a non-empty one

Task 11: Create a Simple GRC Read-Me File with a Text Editor

Use nano to create a short read-me document:

```
nano ~/GRC101_Lab/03_GRC_Evidence_Folder/README.txt
```



The screenshot shows the nano text editor interface. The title bar indicates the file path: `/home/icdfa/GRC101_Lab/03_GRC_Evidence_Folder/README.txt`. The editor content is as follows:

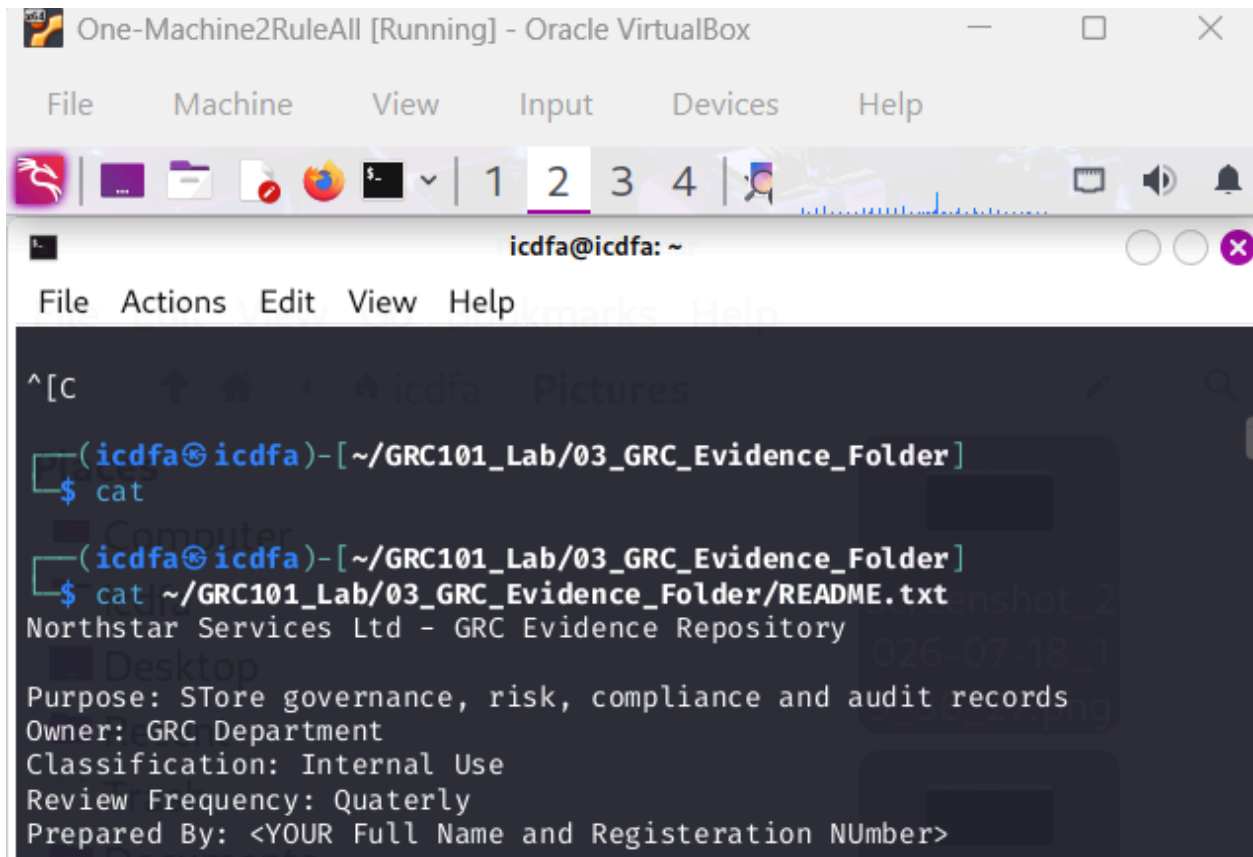
```
GNU nano 8.3
Northstar Services Ltd - GRC Evidence Repository

Purpose: SToRe governance, risk, compliance and audit records
Owner: GRC Department
Classification: Internal Use
Review Frequency: Quaterly
Prepared By: <YOUR Full Name and Registration Number>

# Screenshot_2026-07-18_19_36_27.png
# Screenshot_2026-07-18_19_45_24.png
# Screenshot_2026-07-18_19_52_38.png
# Screenshot_2026-07-18_20_08_24.png
# Screenshot_2026-07-18_20_32_31.png

Other Locations
```

The bottom status bar shows various nano editor shortcuts like `Ctrl+H Help`, `Ctrl+O Write Out`, etc.



The screenshot shows a terminal window titled "One-Machine2RuleAll [Running] - Oracle VirtualBox". The user is logged in as `icdfa@icdfa: ~`. The terminal shows the following commands and output:

```
icdfa@icdfa: ~
File Actions Edit View Help
^[c
(icdfa@icdfa)-[~/GRC101_Lab/03_GRC_Evidence_Folder]
$ cat
(icdfa@icdfa)-[~/GRC101_Lab/03_GRC_Evidence_Folder]
$ cat ~/GRC101_Lab/03_GRC_Evidence_Folder/README.txt
Northstar Services Ltd - GRC Evidence Repository

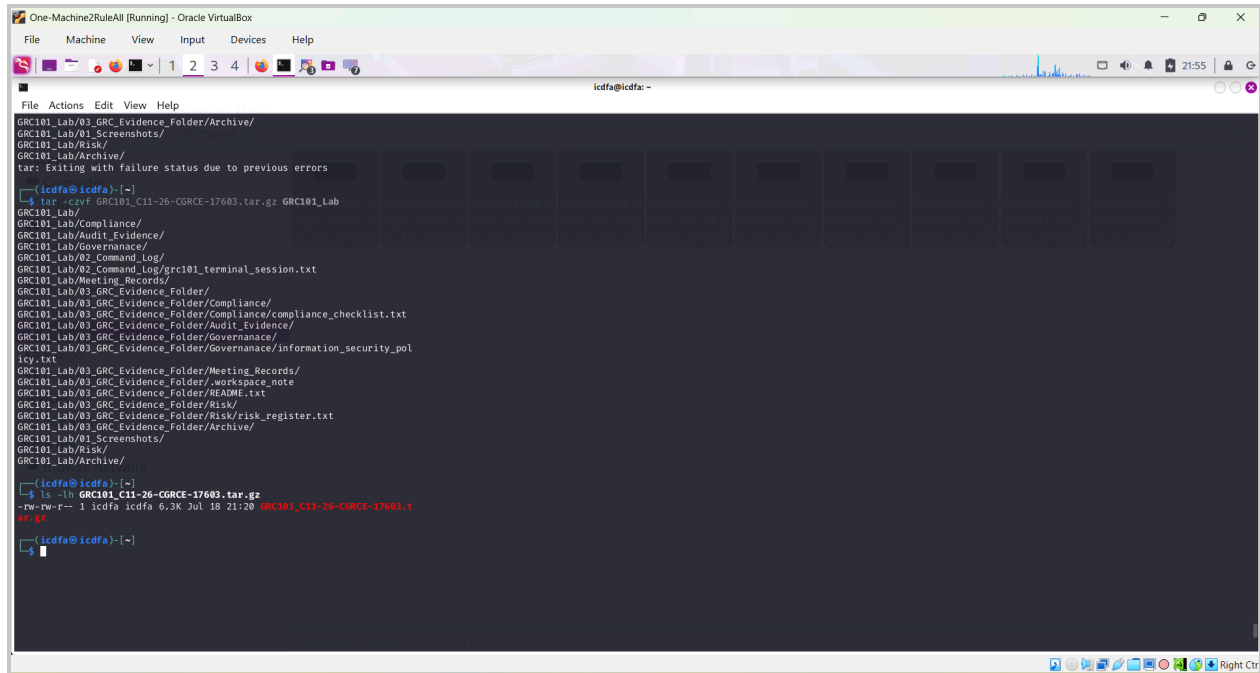
Purpose: SToRe governance, risk, compliance and audit records
Owner: GRC Department
Classification: Internal Use
Review Frequency: Quaterly
Prepared By: <YOUR Full Name and Registration Number>
```

TASK 12: Review the Final Folder Tree and Compress the Work

```
cd ~/GRC101_Lab
```

```
command -v tree && tree 03_GRC_Evidence_Folder
```

```
find 03_GRC_Evidence_Folder -print
```



```
One-Machine2RuleAll [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
icdfa@icdfa: ~
File Actions Edit View Help
GRC101_Lab/03_GRC_Evidence_Folder/Archive/
GRC101_Lab/01_Screenshots/
GRC101_Lab/Risk/
GRC101_Lab/Archive/
tar: Exiting with failure status due to previous errors
icdfa@icdfa: ~
$ tar -czvf GRC101_C11-26-CGRCE-17603.tar.gz GRC101_Lab/
GRC101_Lab/
GRC101_Lab/Compliance/
GRC101_Lab/Audit_Evidence/
GRC101_Lab/Governance/
GRC101_Lab/02_Command_Log/
GRC101_Lab/02_Command_Log/grc101_terminal_session.txt
GRC101_Lab/Meeting_Records/
GRC101_Lab/03_GRC_Evidence_Folder/
GRC101_Lab/03_GRC_Evidence_Folder/Compliance/
GRC101_Lab/03_GRC_Evidence_Folder/Compliance/compliance_checklist.txt
GRC101_Lab/03_GRC_Evidence_Folder/Audit_Evidence/
GRC101_Lab/03_GRC_Evidence_Folder/Governance/
GRC101_Lab/03_GRC_Evidence_Folder/Governance/information_security_policy.txt
GRC101_Lab/03_GRC_Evidence_Folder/Meeting_Records/
GRC101_Lab/03_GRC_Evidence_Folder/workspace_note
GRC101_Lab/03_GRC_Evidence_Folder/README.txt
GRC101_Lab/03_GRC_Evidence_Folder/Risk/
GRC101_Lab/03_GRC_Evidence_Folder/Risk/risk_register.txt
GRC101_Lab/03_GRC_Evidence_Folder/Archive/
GRC101_Lab/01_Screenshots/
GRC101_Lab/Risk/
GRC101_Lab/Archive/
icdfa@icdfa: ~
$ ls -lh GRC101_C11-26-CGRCE-17603.tar.gz
-rw-rw-r-- 1 icdfa icdfa 6.3K Jul 18 21:20 GRC101_C11-26-CGRCE-17603.t
icdfa@icdfa: ~
$
```

Short Reflection Questions

1. Why is a clear folder structure important during an audit?

A clear folder structure makes it easy to locate, verify, and trace evidence quickly, which supports transparency, saves time, and reduces the risk of files being lost, mismatched, or questioned for authenticity during an audit.

2. How can incorrect file names or locations affect compliance evidence?

Incorrect file names or locations can make evidence hard to find or trace back to the right control, raise doubts about the file's authenticity or chain of custody, and in an audit can be treated as if the evidence doesn't exist at all, even if the underlying work was actually done correctly.

3. What is the difference between copying and moving a file?

`cp` creates a duplicate of the file in the new location while keeping the original in place, you end up with two copies.

`mv` transfers the file to the new location and removes it from the original location, you end up with only one copy, just in a different place (`mv` is also used to rename files).

4. Why should administrative access through sudo be limited?

Limiting sudo access reduces the risk of accidental or malicious system changes by ensuring only trusted users can perform privileged actions, which shrinks the attack surface and keeps a clearer, more accountable trail of who made critical changes.

5. How do file permissions support confidentiality and accountability?

Confidentiality:- File permissions restrict who can read, write, or execute a file, ensuring sensitive data is only accessible to authorized users, groups, or roles, preventing unauthorized viewing or leakage.

Accountability:- By tying specific permissions to specific users/groups (owner, group, others), any action taken on a file can be traced back to the individual responsible, making it clear who had the ability to access or change something.

6. What evidence from this lab would you present to show that a GRC folder was created correctly?

To show the GRC folder was created correctly, I'd present:

- A ``ls -li`` output:- showing the folder/file listing with correct names, permissions, ownership, and inode numbers
- B ``stat`` output:- showing creation (Birth), modification, and access timestamps proving when the folder/files were made
- C Ownership details (UID/GID) :- confirming the correct user (icdfa) created and owns the folder
- D Permission settings (e.g. ``rw-rw-r--``) :- proving appropriate access control was applied
- E Folder structure/path: — showing the file sits in the correct, logically named location (e.g. ``~/GRC101_Lab/03_GRC_Evidence_Folder/Governance/``)

Together, this ties the folder's existence, structure, timing, and access control back to a specific accountable user which is exactly what an auditor would look for.

GRC REFLECTION

Explain how organised files, naming, permissions and evidence support governance and compliance.

To show the GRC folder was created correctly, I'd present:

- A `ls -li` output:- showing the folder/file listing with correct names, permissions, ownership, and inode numbers
- B `stat` output:- showing creation (Birth), modification, and access timestamps proving when the folder/files were made
- C Ownership details (UID/GID):- confirming the correct user (icdfa) created and owns the folder
- D Permission settings (e.g. `rw-rw-r--`):- proving appropriate access control was applied
- E Folder structure/path:- showing the file sits in the correct, logically named location (e.g. `~/GRC101_Lab/03_GRC_Evidence_Folder/Governance/`)

Together, this ties the folder's existence, structure, timing, and access control back to a specific accountable user, which is exactly what an auditor would look for.

CHALLENGES AND RESOLUTION

State any problem encountered and how it was resolved.

- A I typed so many commands wrong and got errors as outputs, i had to be more patient with reading and understanding the commands before typing them out
- B I had never seen some of the commands before and was not sure of how they would turn out, i checked them online first to know what to expect
- C Most importantly, I really thought it would be a breeze to finish this assignment, that was a big joke on me. I got better though, so it's fine.

CONCLUSION

Summarise what you learned from the lab.

I really think my Linux skills got better from this single lab, I learned more commands and I know what most of them do now.